

Generation Interconnection System Impact Study Report

For Queue Project Q16-0611

Strahan 138 kV

July 23, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0611. The Interconnection Customer proposes a 78.9 MW solar photovoltaic facility interconnecting at the Strahan 138 kV substation in the Pennsylvania area. The request was received on April 22, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 286 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,052,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Sfpp Lp - Hazleton 138 kV: 133.5% loading (225 MVA) following the outage of the W Campus - Edes Tap 2 Substation 345 kV; DFAX = 27.3%.

Honk Falls - Hazleton 138 kV Line: 122.2% loading (244 MVA) following the outage of the W Campus - Edes Tap 2 345 kV L; DFAX = 27.3%.

Loco Hills - Lewistown 138 kV: 111.9% loading (333 MVA) following the outage of the Edes Tap 2 - West Campus 345 kV; DFAX = 12.9%.

Southbase - Milwaukee 138 kV: 101.8% loading (413 MVA) following the outage of the Fort Humbug - Little Chute 345 kV; DFAX = 7.8%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sfpp Lp - Hazleton 138 kV	Rebuild line with higher-capacity conductor	\$5,338,000	55% (\$2,925,000)
Honk Falls - Hazleton 138 kV Line	Rebuild line with higher-capacity conductor	\$4,117,000	55% (\$2,256,000)
Loco Hills - Lewistown 138 kV	Rebuild line with higher-capacity conductor	\$2,568,000	19% (\$494,000)
Milwaukee - Southbase 138 kV line	Reconductor line	\$4,747,000	29% (\$1,377,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$7,052,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.